

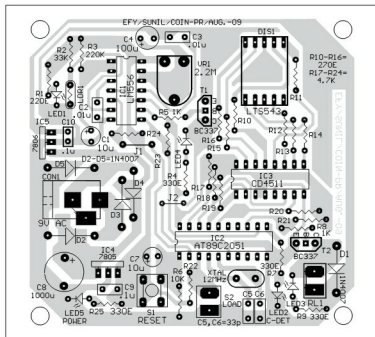


Fig. 4: Component layout for the PCB

After inserting the coin, close load switch S2. Port pin P1.1 of the microcontroller goes high to drive transistor T2 into saturation. Relay RL1 energises and LED3 glows to indicate that the load is now switched on. D1 acts as a free-wheeling diode.

As power is drawn by the load (pin P1.1 high), the count shown on the 7-segment display (DIS1) decrements. Port pin P1.0 of the microcontroller triggers the second timer of NE555. When trigger pin 8 of NE555 goes low, its output pin 9 goes high for a time period decided by preset VR1 and capacitor C4. The high output of the timer is inverted by transistor T1 and fed to port pin P3.2 of the microcontroller (pin 6 of IC2). The count display decrements by '1' after port pin P3.2 of the microcontroller receives five pulses (indicated by glowing of LED4).

Fig. 2 shows the power supply circuit. The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D2 through D5, filtered by capacitor C8 and then regulated by ICs

7805 (IC4) and 7806 (IC5). Capacitors C9 and C10 bypass the ripples present in the regulated 5V and 6V power supplies. LED5 acts as the power-on indicator and resistor R25 limits the current through LED5.

Construction

An actual-size, single-side PCB for the put-coin-and-draw-power circuit is shown in Fig. 3 and its component layout in Fig. 4. Wire the circuit on the PCB and check for proper connections using a multimeter.

Take a 10cm long, hollow, rectangular cuboid of plastic in which a coin can be inserted easily. Drill holes on its opposite sides at the centre and fix LED1 and LDR1 such that these are placed face to face. Align LED1 and LDR1 such that light from LED1 falls on LDR1 directly. Cover the coin-collection assembly to maintain darkness, leaving only coin-insertion and coin-release holes open. Extend LED1 and LDR1 from the PCB to the coin assembly using wires.

When a coin is inserted into the coin-collection assembly, light falling on the LDR is blocked as the coin

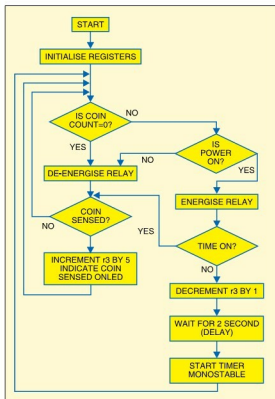


Fig. 5: Flow-chart of the microcontroller program

comes in between LED1 and LDR1. LDR1 offers high resistance in the absence of light and this is detected by one timer of NE556 and indicated by LED2.

You can insert up to nine coins and this count is displayed on DIS1. When you close the load switch S2, the relay energises and the load turns on. The count displayed on DIS1 starts decrementing after LED4 blinks five times. The relay de-energises to turn the load off when the count displayed on DIS1 becomes zero. Load-'on' duration can be adjusted using preset VR1. Never insert the coin without pressing reset switch S1.

Software

The source program is written in Assembly language and assembled using Metalink's ASM51 assembler, which is freely available on the Internet for download. The source program has been well commented for easy understanding. It works as per the flow-chart shown in Fig. 5.